

Heart Disease MLOps Project

Reproducible model development, experiment tracking, API delivery and operational evidence

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303

UCI Cleveland records

13

patient input features

2

cross-validated classifiers

Executive summary

This submission implements a complete heart-disease classification workflow: data acquisition and EDA, leakage-aware preprocessing, model comparison and MLflow tracking, a packaged scikit-learn pipeline, a tested FastAPI service, CI automation, Docker packaging, Kubernetes deployment manifests, request logging and Prometheus metrics.

The selected Logistic Regression model achieved mean five-fold cross-validation ROC-AUC **0.9082**. Its held-out ROC-AUC was **0.9632** on the fixed stratified split. Docker build, two-replica Docker Desktop Kubernetes rollout, live API checks and five local tests passed; deployment evidence is included in this report.

Safety: This is an educational demonstration, not a medical device. It is not validated for diagnosis, triage or treatment and must not be used for clinical decisions.

Project: github.com/priyansh1114/MLOps · **Run environment:** Windows, Docker Desktop Kubernetes

Dataset and data quality

The project uses the processed Cleveland subset of the UCI Heart Disease dataset.

`src/heart_disease_mlop/data_pipeline.py` downloads the UCI source file when the local snapshot is absent; the checked-in `data/heart.csv` makes routine runs independent of network access. Original target values 1–4 map to disease-present class 1; value 0 remains class 0.

303

rows

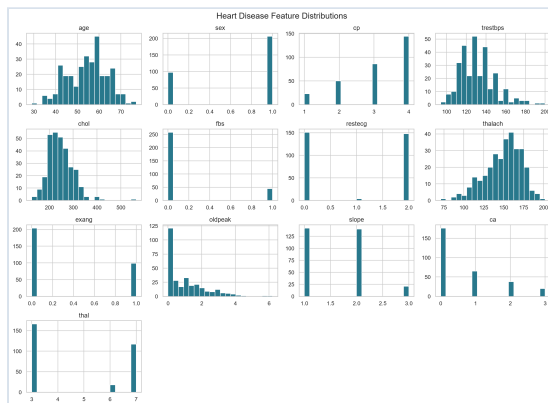
13

input features

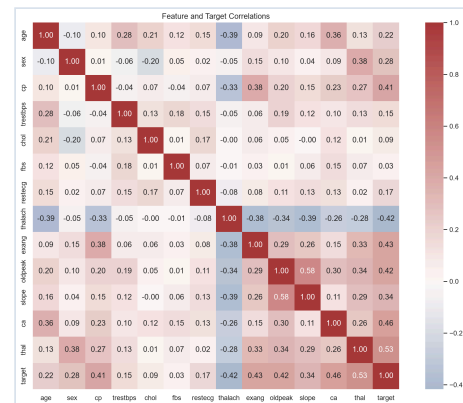
164 / 139

class 0 / class 1

Question marks are parsed as missing values. Four missing `ca` values and two missing `thal` values are preserved for training-only imputation. This avoids leakage from computing imputation statistics on the held-out partition. Feature histograms, feature/target correlations and the class balance chart are generated by the EDA command.



Feature distribution histograms



Feature and target correlation heatmap



Target balance: 164 no-disease and 139 disease-present records

Generated EDA summary: `artifacts/eda/summary.json`. Source: [UCI processed Cleveland data](#).

One fitted pipeline from training to inference

The input schema is fixed in `FEATURE_COLUMNS`. Numeric inputs use median imputation and standardization; coded categories use most-frequent imputation and one-hot encoding with unknown-category tolerance. Transformers are fitted within cross-validation folds and saved with the final estimator.

Feature group	Columns	Training transformation
Continuous / measured	age, trestbps, chol, fbs, thalach, exang, oldpeak, ca	Median imputation; standard scaling
Coded categories	sex, cp, restecg, slope, thal	Most-frequent imputation; one-hot encoding
Target	target	0 = no disease; original 1–4 mapped to 1

Libraries, methods, and verified environment

Libraries (verified versions)	Methods used
Python 3.12 stdlib; pandas 3.0.6; NumPy 2.5.3	<code>Path</code> , <code>urlretrieve</code> , <code>read_csv(na_values="?)</code> , <code>to_numeric</code> ; acquisition, cleaning and tabular data
scikit-learn 1.9.1	<code>ColumnTransformer/Pipeline</code> ; imputers, scaler and one-hot encoder; stratified split, <code>GridSearchCV</code> , <code>StratifiedKFold</code> ; Logistic Regression, Random Forest; accuracy, precision, recall, F1, ROC-AUC and diagnostic displays
MLflow 3.16.1; joblib 1.6.0	Nested runs, <code>log_params/log_metrics/log_artifacts</code> ; <code>joblib.dump/load</code> pipelines
Matplotlib 3.11.2; Seaborn 0.13.2	Histograms, heatmap, barplot, diagnostic figures and <code>savefig</code>
FastAPI 0.142.2; Pydantic 2.13.5; Uvicorn 0.54.0	GET/POST routes, HTTP middleware, <code>BaseModel/Field</code> validation, ASGI server
Prometheus client 0.26.0; pytest 9.1.1; Ruff 0.16.10	<code>Metric Counter/Histogram</code> ; API/data tests; E4/E7/E9/F lint

Versions above were inspected in the local Python 3.12 project environment; `requirements.txt` intentionally specifies supported minimums, not exact pins.

Reproduction: install `requirements.txt`, then run `python -m src.heart_disease_mlop.eda` and `python -m src.heart_disease_mlop.train_model`. Run the API with `uvicorn api.main:app`. The selected full pipeline is saved to `artifacts/best_model.joblib`.

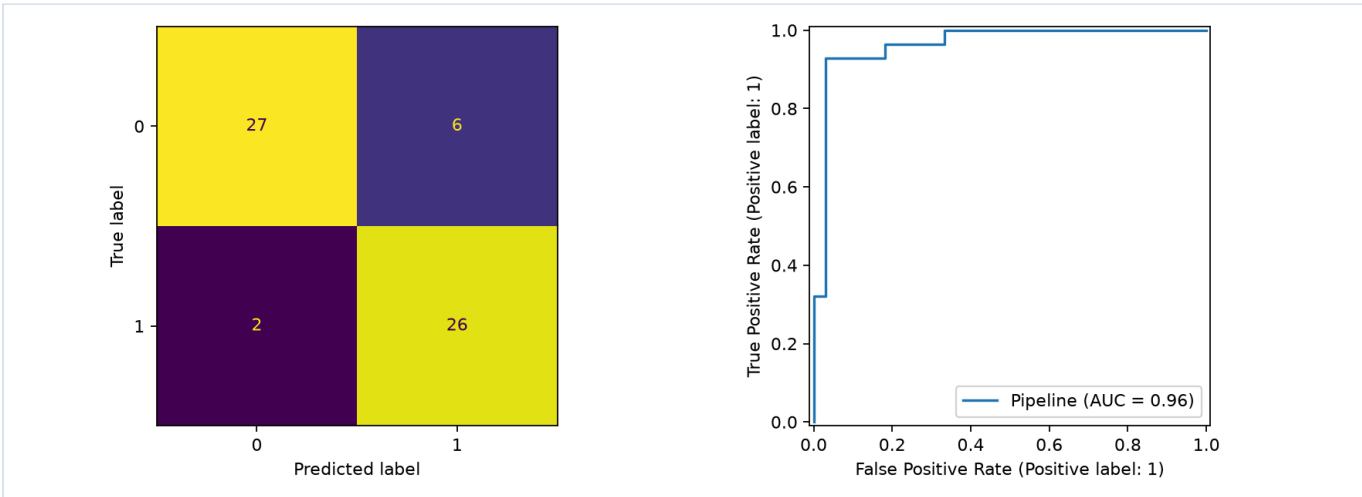
Artifact trust: joblib/pickle can execute code when loaded. Only load trusted project artifacts.

Two models, stratified five-fold CV

The dataset was split 80/20 with stratification and random seed 42. Grid search ran only on the training partition; mean CV ROC-AUC selected the final model. Accuracy, precision, recall and ROC-AUC were recorded for both CV and the independent holdout partition.

Model / best parameters	CV ROC-AUC mean ± SD	CV accuracy	CV precision	CV recall	Holdout accuracy	Holdout precision	Holdout recall	Holdout ROC-AUC
Logistic Regression C=0.1	0.9082 ± 0.0199	0.8386	0.8506	0.7921	0.8689	0.8125	0.9286	0.9632
Random Forest depth=None, leaf=2, trees=100	0.9001 ± 0.0217	0.8181	0.8186	0.7826	0.8689	0.8125	0.9286	0.9448

Logistic Regression is selected because it has the higher mean cross-validated ROC-AUC. The holdout metrics are descriptive for this small split; they are not used to choose hyperparameters. Both models use class weighting. The tested Logistic Regression grid was C = 0.1, 1, 10; Random Forest grid searched 100/200 trees, unlimited/depth-6 trees and leaf sizes 1/2.



Selected model: holdout confusion matrix and ROC curve

Full grid results and selected parameters: `artifacts/evaluation/` and `artifacts/training_summary.json`. No external validation cohort, calibration analysis or subgroup fairness results are claimed.

MLflow runs and recorded outputs

Training creates the local MLflow experiment `heart_disease_prediction`. A parent run records the selected model; nested runs compare Logistic Regression and Random Forest. Parameters, CV metric means and standard deviations, holdout metrics, model artifacts, search tables and diagnostic plots are logged per model.

Tracked item	Recorded evidence	Repository artifact
Parameters	Model type, split seed/size, fold count, best grid parameters, selection metric	<code>mlruns/</code>
Metrics	CV mean and standard deviation; holdout accuracy, precision, recall, F1 and ROC-AUC	<code>artifacts/training_summary.json</code>
Models	Fitted preprocessing + estimator pipelines	<code>artifacts/best_model.joblib</code> , per-model joblib files
Plots and tables	Grid-search rows, confusion matrix and ROC curve	<code>artifacts/evaluation/</code>

Model registry / local UI

The local tracking URI points to the repository's `mlruns/` store. Inspect runs with:

```
$env:MLFLOW_ALLOW_FILE_STORE = "true"
mlflow ui --backend-store-uri .\mlruns --host 127.0.0.1 --port 5000
```

The Docker image includes the saved final model and loads it at service startup. If that artifact is absent during a development run, the API can train a replacement from the local dataset. For production, pin and version the model together with a dependency lock, dataset checksum and MLflow run ID.

Reproducibility boundary: fixed seeds and checked-in data/artifacts support repeatable local evaluation. Requirements are lower bounds, not a lockfile; platform/library updates may change exact fitted bytes and scores.

Validated FastAPI prediction API

Route	Purpose	Verification
GET /health	Process and model readiness	HTTP 200; model_loaded=true
POST /predict	13-feature validated patient payload	Prediction, class-1 probability and predicted-class confidence
GET /metrics	Prometheus counters and latency histogram	HTTP 200; request metrics observed
GET /docs	Interactive OpenAPI / Swagger documentation	Live browser evidence on page 8

Example request

```
POST /predict
Content-Type: application/json

{"age":52,"sex":1,"cp":0,"trestbps":125,"chol":212,
"fbs":0,"restecg":1,"thalach":168,"exang":0,"oldpeak":1.0,
"slope":2,"ca":2,"thal":2}
```

The response shape is `{"prediction":1,"probability":0.7242,"confidence":0.7242}` for this test request in the built container. The response rounds probability fields to four decimals. Pydantic constrains coded categories and rejects negative values; requests do not include raw feature values in logs.

Local access after Docker start

```
docker run --name heart-disease-api-local -p 8000:8000 heart-disease-api:latest
curl http://127.0.0.1:8000/health
curl http://127.0.0.1:8000/docs
```

The screenshot evidence folder contains a captured Swagger page, readiness JSON and Prometheus metrics response from the running Docker service. The live service is currently published on localhost port 8000.

The predicted label and score are model outputs only. A probability from this model is not a clinically calibrated individual risk estimate.

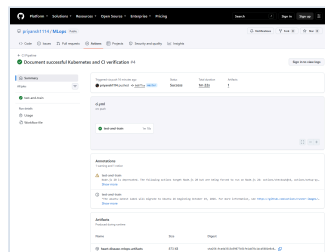
Quality gates and local verification

`.github/workflows/ci.yml` runs on pushes and pull requests to `main` and `master`. It installs the requirements, runs Ruff, runs pytest, trains both model families, generates EDA outputs and uploads artifacts even if an earlier job step fails.

Checkout → Python 3.12 + dependencies → Ruff
→ pytest → training + MLflow → EDA → workflow artifacts

Executed locally for this deliverable

Check	Result	Scope
<code>python -m pytest -q</code>	PASS · 5 passed	API health, prediction, metrics and data/preprocessor tests
<code>python -m ruff check --select E4,E7,E9,F src api tests</code>	PASS · All checks passed	Source, API and tests
Docker image build	PASS · image built	<code>heart-disease-api:latest</code>
Docker health / prediction / metrics	PASS · HTTP 200	Running container on localhost:8000
GitHub Actions run #7	PASS · all steps	Workflow run · download artifact



Hosted CI run #4: lint, tests, model training, EDA and artifact upload all succeeded.

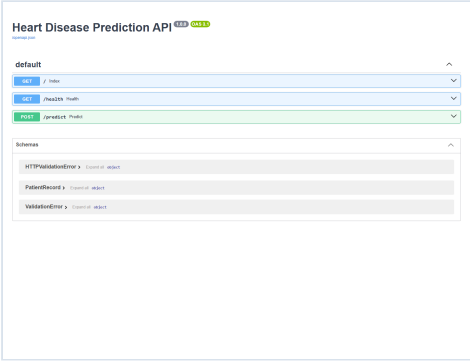
The initial hosted run exposed missing pytest imports; `pytest.ini` now adds the repository root and `src` to Python's import path. The latest successful hosted run emitted a runner/action Node-version deprecation notice; it did not fail the workflow.

Reproduce the quality gates from the project root with the Ruff command above followed by `python -m pytest -q`. Hosted run #7 uploaded the [heart-disease-mlops-artifacts](#) bundle. The included screenshot records a prior successful hosted run #4.

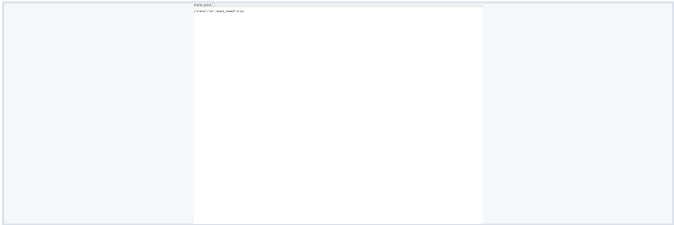
Docker verification evidence

The Docker image was built successfully from the repository's Python 3.12 slim base image using the declared requirements. Docker Desktop's Linux engine is running; the container remains up and is published on port 8000.

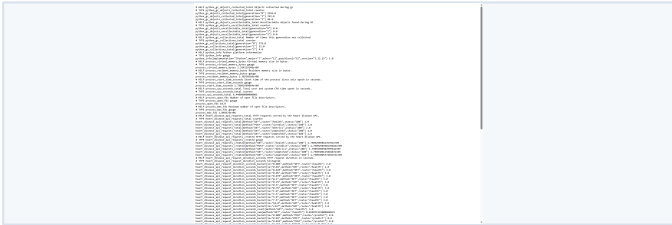
Build / runtime evidence	Observed result
Image tag	heart-disease-api:latest
Image ID	sha256:91ad6ca68e286581a95a3bb3aba233d26e6096ae1fb5db3c4aa3bd2f336eef65
Container name / status	heart-disease-api-local · Up, host 0.0.0.0:8000 → container 8000
/health	{"status":"ok","model_loaded":true}
Sample /predict	HTTP 200 · prediction 1 · probability 0.7242 · confidence 0.7242
/metrics	HTTP 200 · request counter and duration histogram exposed



Docker-container Swagger UI with the prediction and health routes.



Docker /health response.



Docker /metrics: counters and latency histogram.

Build context excludes the local virtual environment, Git data, MLflow run store and caches via `.dockerignore`.

Request observability and cluster status

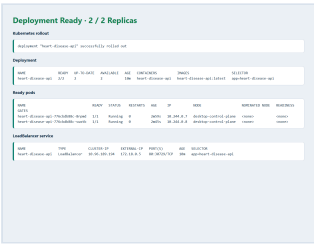
FastAPI middleware logs request method, route, status and duration without patient features. Prometheus `/metrics` exposes request counters and latency histograms. Kubernetes provides two replicas, health probes, scrape annotations and a LoadBalancer Service.

```
heart_disease_api_requests_total{method="POST",route="/predict",status="200"} 1.0
heart_disease_api_request_duration_seconds_count{method="POST",route="/predict"} 1.0
```

Deployment evidence boundary

Target	Status	Evidence / blocker
Docker	Verified	Health, prediction, metrics
Kubernetes	2/2 ready	LoadBalancer Service; health and prediction verified
Monitoring	Verified	Request logs, Prometheus counter and latency histogram
Public cloud URL	Not claimed	Local cluster service only

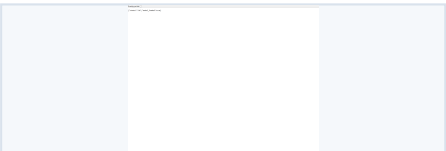
The helper script imports the image into Docker Desktop's separate containerd store; the manifest uses `imagePullPolicy: Never`. Kubernetes was verified through `kubectl port-forward`. Its LoadBalancer address is cluster-local, not public.



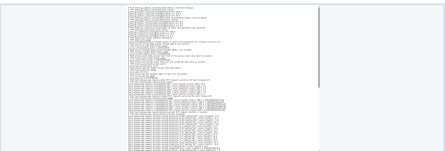
Rollout, two ready pods and LoadBalancer Service.



Cluster Swagger UI.



Cluster health check.



Cluster Prometheus metrics.

The local Docker Desktop Kubernetes cluster satisfies the assignment's local-cluster deployment option. No public cloud URL is claimed.

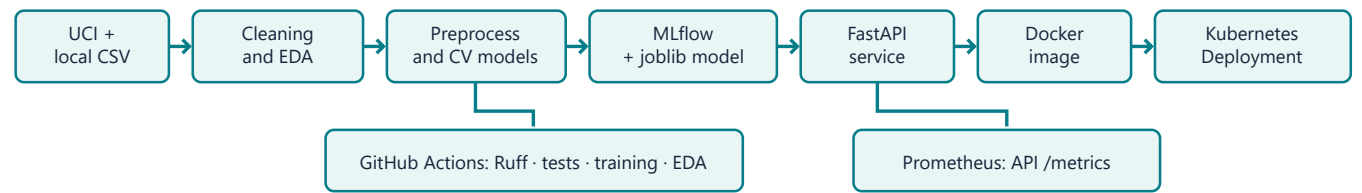
Coverage summary and next actions

Assignment area	Coverage	Repository artifact / evidence
1. Data acquisition & EDA	Implemented	Download-on-missing, data snapshot, histograms, heatmap, balance and summary
2. Feature engineering & modeling	Implemented	Leakage-aware preprocessing, two grid-searched models, five-fold CV and metrics
3. Experiment tracking	Implemented	Nested MLflow runs with parameters, metrics, artifacts and plots
4. Packaging & reproducibility	Implemented	Serialized full pipeline, requirements, inference command
5. CI/CD & testing	Implemented / hosted run passed	GitHub Actions #7, Ruff, 5 pytest tests, training and artifact upload
6. Model containerization	Built, started & verified	Dockerfile, live API and captured screenshots
7. Production deployment	Local Kubernetes deployed	Two ready replicas; Service and API verified; no public cloud URL
8. Monitoring & logging	Implemented / verified	Privacy-aware request logs and Prometheus counters/histogram
9. Documentation & reporting	PDF + Markdown	This 10-page PDF, report.md, README, EDA and API evidence

Setup and installation

From the repository root: `py -3.12 -m venv .venv → .\.venv\Scripts\Activate.ps1 → pip install -r requirements.txt`. Then run the EDA and training commands on page 3.

Architecture



Deliverable coverage

This report maps each of the nine written and technical assignment areas to its implementation and evidence, including the code, dataset, preprocessing/training/inference scripts, tests, CI workflow, container, Kubernetes manifest, monitoring endpoint, screenshots and 10-page PDF.

Limitations: the historical dataset is small; there is no external validation, calibration/fairness/drift assessment or clinical review. The local Kubernetes LoadBalancer is not a public-cloud deployment. This model must not be used for real medical decisions.

Key deliverables: source in `src/` and `api/`; data and EDA/model artifacts under `data/` and `artifacts/`; tests in `tests/` with `pytest.ini`; CI at `.github/workflows/ci.yml`; Dockerfile, `k8s/deployment.yaml` and `scripts/load_docker_desktop_image.ps1`; evidence in `screenshots/`.